MAT 3007 – Optimization

The Gradient Descent Method

Lecture 23

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Recap

Optimization Methods for One-Dimensional Problems:

- Bisection method: solve $f'(x) = 0$.
- Golden section method.

Higher Dimensional Problems:

- Generate iterates of the form: $\boldsymbol{x}^{k+1} = \boldsymbol{x}^k + \alpha_k \boldsymbol{d}^k$.
- $\boldsymbol{d}$ is a descent direction of f at $\boldsymbol{x}$ if $\nabla f(\boldsymbol{x})^\top \boldsymbol{d} < 0$.
- Taking a small step along a descent direction reduces the objective function value: there exists $\epsilon > 0$ such that

$$f(\boldsymbol{x} + \alpha \boldsymbol{d}) < f(\boldsymbol{x}) \quad \forall\ \alpha \in (0, \epsilon].$$

Bisection Method

Bisection method uses the idea that the local minimizer must satisfy the first-order necessary conditions: $f'(x) = 0$.

Therefore, the problem becomes a root-finding problem for

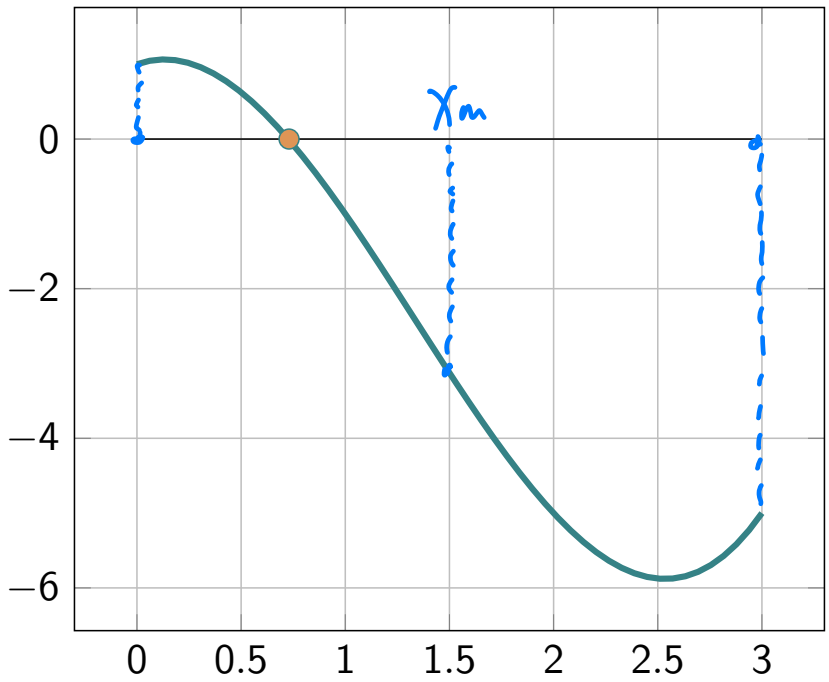
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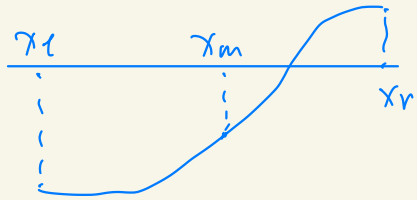
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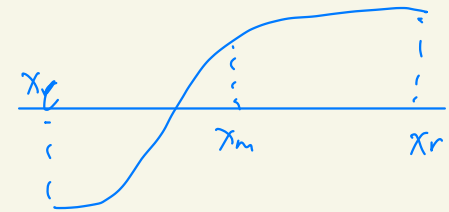
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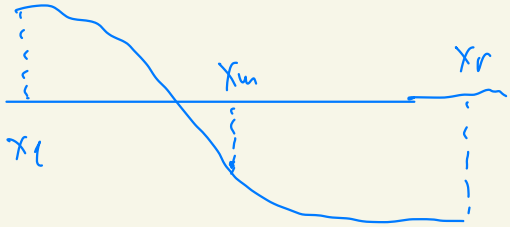




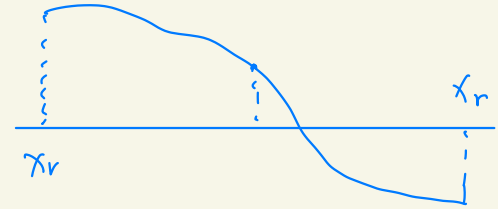
root is in (x_m, x_r)



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Root Finding Algorithm: Bisection Method

Assume we can find x_ℓ and x_r such that $g(x_\ell) < 0$ and $g(x_r) > 0$.

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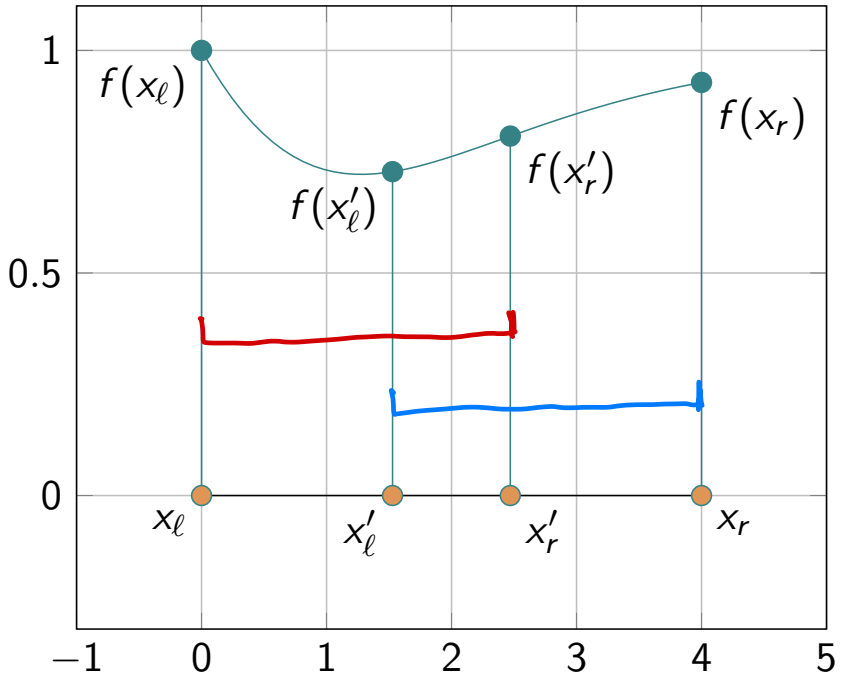
1. Define $x_m = \frac{x_\ell + x_r}{2}$.
2. If $g(x_m) = 0$, then output x_m.
3. Otherwise:
   - If $g(x_m) > 0$, then let $x_r = x_m$.
   - If $g(x_m) < 0$, then let $x_\ell = x_m$.
4. If $|x_r - x_\ell| < \varepsilon$: stop and output $\frac{x_\ell + x_r}{2}$, otherwise go back to step 1.

- To find a point that is ϵ-close to a stationary point, we need at most $\log_2(\frac{x_r - x_\ell}{\varepsilon})$ iterations.

Golden Section Method

Assume f has a unique local minimizer x^* in the range $[x_\ell, x_r]$, then we have a very efficient way to find x^*:

- We call f unimodal if it has a single local minimizer or local maximizer.



Golden Section Method

Assume we start with $[x_\ell, x_r]$. Assume $0 < \phi < 0.5$.

1. Set $x'_\ell = \phi x_r + (1-\phi)x_\ell$ and $x'_r = (1-\phi)x_r + \phi x_\ell$.
2. If $f(x'_\ell) < f(x'_r)$, then the minimizer must lie in $[x_\ell, x'_r]$, so set $x_r = x'_r$.
3. Otherwise, the minimizer must lie in $[x'_\ell, x_r]$, so set $x_\ell = x'_\ell$.
4. If $x_r - x_\ell < \varepsilon$, output $\frac{x_\ell + x_r}{2}$, otherwise go back to step 1.

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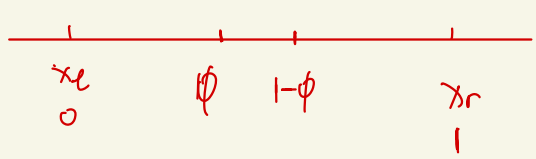
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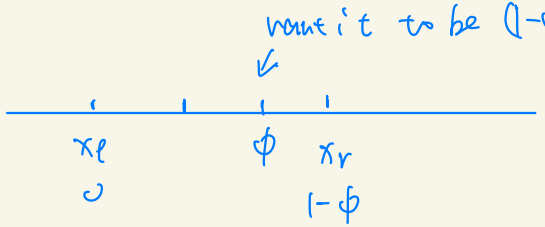
$$\phi = \frac{3-\sqrt{5}}{2} \quad \text{and} \quad 1-\phi = \frac{\sqrt{5}-1}{2} = 0.618.$$

How to choose ϕ.



iteration 1

suppose $f(1-\phi) > f(\phi)$, then we update $1-\phi$ to be the new right bound



iteration 2

we want $(1-\phi)^2 = \phi$, solve ϕ

Abstract Descent Method: A First Scheme

Schematic Descent Directions Method

1. Initialization: Select an initial point $\boldsymbol{x}^0 \in \mathbb{R}^n$.

For $k = 0, 1, \ldots$:

2. If a stopping criterion holds, then STOP and $\boldsymbol{x}^k$ is the output.
3. Pick a descent direction $\boldsymbol{d}^k$.
4. Find a step size α_k satisfying $f(\boldsymbol{x}^k + \alpha_k \boldsymbol{d}^k) < f(\boldsymbol{x}^k)$.
5. Set $\boldsymbol{x}^{k+1} = \boldsymbol{x}^k + \alpha_k \boldsymbol{d}^k$.

- In gradient descent method, we choose $\boldsymbol{d}^k = -\nabla f(\boldsymbol{x}^k)$.
- A stopping criterion: $\|\nabla f(\boldsymbol{x}^k)\| \leq \varepsilon$ with tolerance $\varepsilon > 0$.
- Step sizes options: constant stepsize $\alpha_k = \bar{\alpha}$ and the exact line search $\alpha_k = \arg\min_{\alpha \geq 0} f(\boldsymbol{x}^k + \alpha \boldsymbol{d}^k)$.

More Step Size Strategies

Backtracking / Armijo Line Search

Let $\boldsymbol{d}^k$ be a descent direction. With $\gamma \in (0,1)$, find α_k such that

$$f(\boldsymbol{x}^k + \alpha_k \boldsymbol{d}^k) - f(\boldsymbol{x}^k) \leq \gamma \alpha_k \cdot \nabla f(\boldsymbol{x}^k)^\top \boldsymbol{d}^k.$$

- This condition is called Armijo condition.

< 0 because d^k is a descent direction.

If we choose $d^k = -\nabla f(x^k)$,

$$f(x^k + \alpha_k d^k) - f(x^k) \leq -\gamma \alpha_k \|\nabla f(x^k)\|^2$$

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Procedure: (Backtracking / Armijo Line Search)

1. Start with $\alpha = 1$ and pick $\sigma \in (0,1)$.
2. If $f(\boldsymbol{x}^k + \alpha \boldsymbol{d}^k) \leq f(\boldsymbol{x}^k) + \gamma \alpha \cdot \nabla f(\boldsymbol{x}^k)^\top \boldsymbol{d}^k$, choose $\alpha_k = \alpha$. Otherwise, set $\alpha = \sigma \alpha$ and repeat this step.

we try $\alpha = 1, \sigma, \sigma^2, \sigma^3 \ldots$ until it satisfies Armijo condition.

Armijo Line Search: Discussion

Why does this work?

- By Taylor expansion, if α is sufficiently small, we have

$$f(\boldsymbol{x}^k+\alpha\boldsymbol{d}^k) = f(\boldsymbol{x}^k)+\alpha\nabla f(\boldsymbol{x}^k)^\top \boldsymbol{d}^k+o(\alpha) < f(\boldsymbol{x}^k)+\gamma\alpha\cdot\nabla f(\boldsymbol{x}^k)^\top \boldsymbol{d}^k.$$

  Therefore, as long as α is small enough, the Armijo condition must be satisfied.

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Illustration:

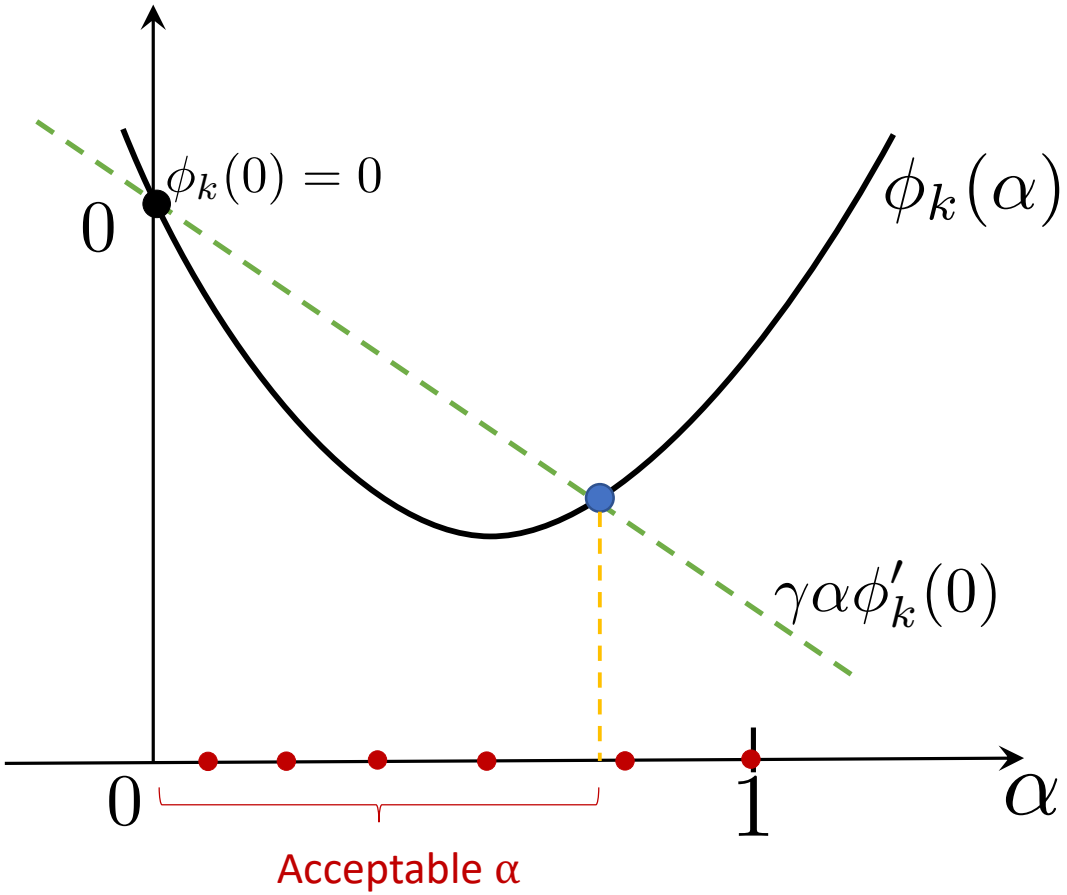
- Define $\phi_k(\alpha) := f(\boldsymbol{x}^k + \alpha\boldsymbol{d}^k) - f(\boldsymbol{x}^k)$. Then, we have

$$\phi_k'(\alpha) = \nabla f(\boldsymbol{x}^k + \alpha\boldsymbol{d}^k)^\top \boldsymbol{d}^k, \quad \phi_k'(0) = \nabla f(\boldsymbol{x}^k)^\top \boldsymbol{d}^k.$$

- The Armijo condition is then equivalent to:

$$\textit{find } \alpha \textit{ with } \phi_k(\alpha) \le \gamma\alpha\cdot\phi_k'(0).$$

Armijo Line Search: Visualization



- The green line $\gamma\alpha\phi_k'(0)$ must be above $\phi_k(\alpha)$ for small enough α, as guaranteed by Taylor expansion.

Other Step Size Strategies

Diminishing Step Sizes:

- Choose $\{\alpha_k\}_k$ with $\alpha_k \to 0$ and

$$\sum_{k=0}^{\infty} \alpha_k = \infty.$$

- Such a choice is popular and important for stochastic optimization methods. For instance, we can choose $\alpha_k = 1/(k+1)$.

Many more: Barzilai-Borwein step sizes, non-monotone strategies, and some pupular step sizes in mahcin learning, such as Adam, AdaGrad ...

The Full Gradient Method and Numerical Examples

Gradient Descent Method

1. Initialization: Select an initial point $\boldsymbol{x}^0 \in \mathbb{R}^n$.

For $k = 0, 1, \ldots$:

2. If $\|\nabla f(\boldsymbol{x}^k)\| \leq \varepsilon$, then STOP and $\boldsymbol{x}^k$ is the output.
3. Pick a step size α_k. For instance, by a line search procedure (exact line search or backtracking) on the function

$$\phi(\alpha) = f(\boldsymbol{x}^k - \alpha \nabla f(\boldsymbol{x}^k)).$$

4. Set $\boldsymbol{x}^{k+1} = \boldsymbol{x}^k - \alpha_k \nabla f(\boldsymbol{x}^k)$.

Example: A Quadratic Problem

Test the methods and solve the simple problem:

$$\min_{\boldsymbol{x}} \; f(\boldsymbol{x}) = x_1^2 + 2x_2^2 = \frac{1}{2}\boldsymbol{x}^\top \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix} \boldsymbol{x}$$

with the step size strategies:

- Exact line search (EL).
- Backtracking / Armijo line search (AL).

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We use the initial point $\boldsymbol{x}^0 = (2, 1)^\top$, $\texttt{tol} = 10^{-5}$, $\sigma = \frac{1}{2}$, $\gamma = 0.1$.

- **EL:** The method stops after 13 iterations with a solution that is very close the optimal value $\boldsymbol{x} = 10^{-5} \cdot (0.1254, -0.0627)^\top$.
- **AL:** We only need 2 iterations to reach a similar solution.

Example: Gradient Method for Quadratic Functions



```
function [x,obj] = gm_quadratic(A,b,x0,eps)

x       = x0;       iter    = 0;
g       = A*x + b;  ng      = norm(g);

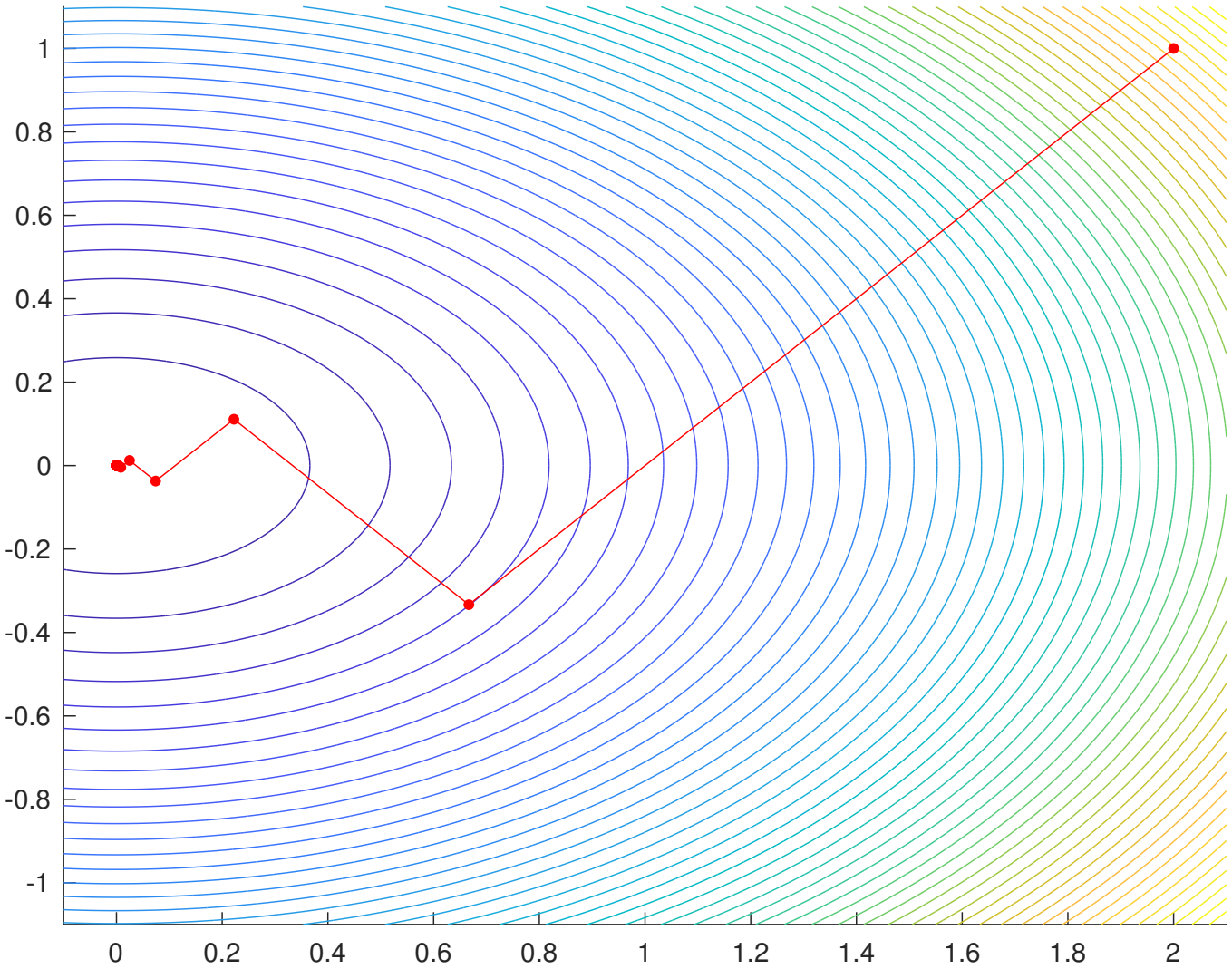
fprintf(1,'- - - grad. method ; n = %g\n',length(b));
fprintf(1,'ITER ; OBJ.VAL ; G.NORM ; STEP.SIZE\n');

while ng > eps && iter < 10000
 iter    = iter + 1;
 alpha   = ng^2 / (g'*A*g);
 x       = x - alpha*g;
 g       = A*x + b;
 ng      = norm(g);
 obj     = 0.5*x'*A*x + b'*x;
 fprintf(1,'[%4i] ; %2.6f ; %2.6f ; %1.2f\n',iter,obj,
     ng,alpha);
end
```

Example: Gradient Method with Backtracking

```
1  function [x,obj] = gm_quadratic_armijo(A,b,x0,eps)
2
3  x       = x0;       iter    = 0;
4  g       = A*x + b;  obj     = 0.5*x'*A*x + b'*x;
5  ng      = norm(g);  gamma   = 1e-1;
6
7  while ng > eps && iter < 10000
8   iter    = iter + 1;
9
10  alpha   = 1;                          xn      = x - alpha*g;
11  objn    = 0.5*xn'*A*xn + b'*xn;  r       = -gamma*ng^2;
12
13  while objn - obj > r*alpha
14      alpha   = 0.5*alpha;          xn      = x - alpha*g;
15      objn    = 0.5*xn'*A*xn + b'*xn;
16  end
17  x = xn; g = A*x + b; ng = norm(g); obj = objn;
18 end
```

Illustration: Iteration Path for Exact Line Search



Perpendicular Steps

The gradient descent method with exact line search follows a zig-zag path towards the solution.

The directions between two consecutive steps are perpendicular:

$$(\boldsymbol{d}^{k+1})^\top \boldsymbol{d}^k = 0.$$

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Why?

- If α_k is the minimizer of $\phi(\alpha) = f(\boldsymbol{x}^k + \alpha \boldsymbol{d}^k)$. By FONC, we have $\phi'(\alpha_k) = 0$, which means:

EL: $\min_{\alpha \in [0,\infty)} f(x^k + \alpha d^k)$

If α_k is a solution, then $\alpha_k > 0$ because d^k is a descent direction and there must exist positive α s.t $f(x^k + \alpha d^k) < f(x^k)$. $\Rightarrow \alpha_k$ is in the interior of $[0,\infty)$

$\Rightarrow$ By FONC, $\phi'(\alpha_k) = 0$

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$$0 = \phi'(\alpha_k) = \nabla f(\boldsymbol{x}^k + \alpha_k \boldsymbol{d}^k)^\top \boldsymbol{d}^k = -(\boldsymbol{d}^{k+1})^\top \boldsymbol{d}^k.$$

Example: A Nonlinear Problem

We consider the nonlinear simple problem:

$$\min_{\boldsymbol{x}} \; f(\boldsymbol{x}) = \exp(x_1 + x_2) + x_1^2 + 3x_2^2 - x_1 x_2$$

and test the step size strategies:

- Exact line search.
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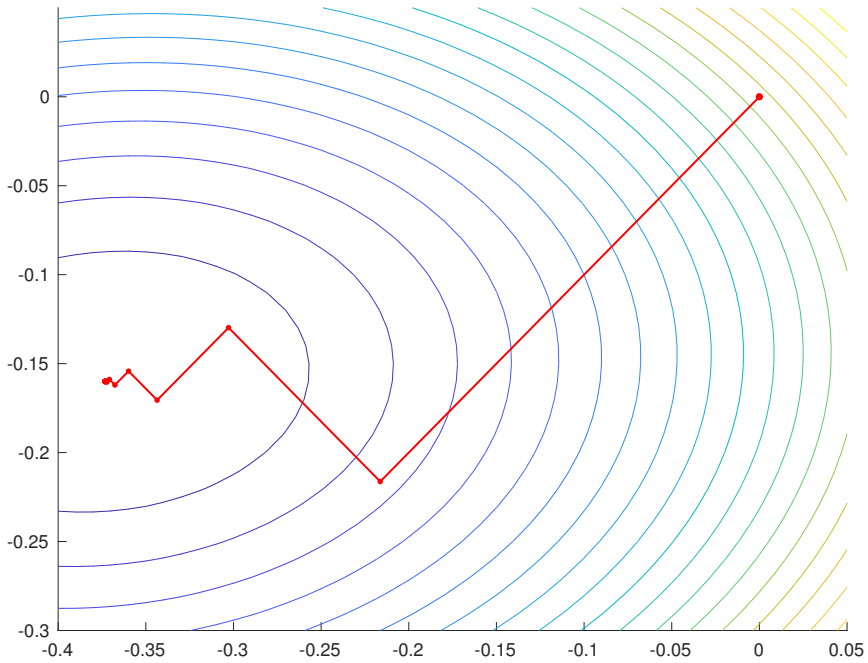
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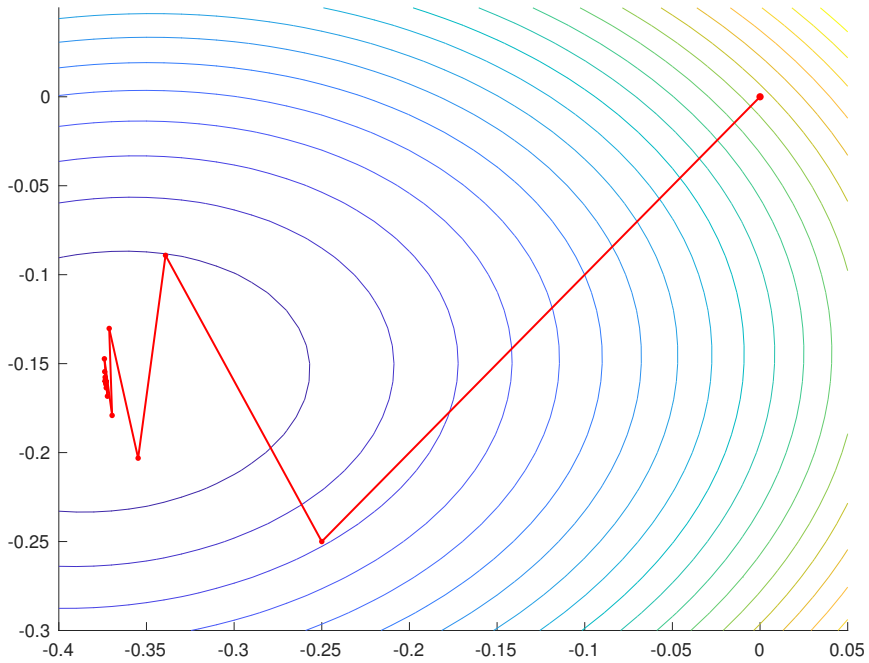
We use the initial point $\boldsymbol{x}^0 = (0, 0)^\top$, $\texttt{tol} = 10^{-6}$, $\sigma = 0.5$, and $\gamma = 0.1$.

- **EL:** The method stops after 19 iterations.
- **AL:** We require 35 iterations to reach a similar solution. With $\gamma = 0.5$ only 18 iterations are needed.

Illustration: Solution Path



EL



AL

Gradient Method: Convergence and Properties

Convergence of the Gradient Method

We are interested in the convergence of the gradient method.

- We study the behavior of $\{f(\boldsymbol{x}^k)\}_k$, $\{\|\nabla f(\boldsymbol{x}^k)\|\}_k$, $\{\boldsymbol{x}^k\}_k$, ...

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Global and Local Convergence:

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Rate of Convergence:

- We are interested in how fast an algorithm converges.
- Under different assumptions, we may establish different rates of convergence.

Accumulation Points

Definition: Accumulation Point

A point $\boldsymbol{x}$ is an accumulation point of $\{\boldsymbol{x}^k\}_k$ if for every $\varepsilon > 0$, there are infinitely many numbers k with $\boldsymbol{x}^k \in B_\varepsilon(\boldsymbol{x})$.

We continue we several remarks:

- If $\boldsymbol{x}$ is an accumulation point of $\{\boldsymbol{x}^k\}_k$ then there is a subsequence $\{\boldsymbol{x}^{k_\ell}\}_\ell$ that converges to $\boldsymbol{x}$.

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- If $\{\boldsymbol{x}^k\}_k$ converges to some $\boldsymbol{x} \in \mathbb{R}^n$, then $\boldsymbol{x}$ is the unique accumulation point of $\{\boldsymbol{x}^k\}_k$.
- A bounded sequence always possesses at least one accumulation point.

Examples: Accumulation Points

Examples:

- The sequence $\{a_k\}_k$ with $a_k = (-1)^k$ has the two accumulation points $a = +1$ and $a = -1$.

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- The sequence $\{a_k\}_k$ with $a_k = (-1)^k$ has the two accumulation points $a = +1$ and $a = -1$.
- The sequence

$$a_k := \begin{cases} k & k \text{ is odd}, \\ 0 & k \text{ is even}, \end{cases}$$

is not bounded. However, it has the accumulation point $a = 0$.

Theorem: Global Convergence

Let $f : \mathbb{R}^n \to \mathbb{R}$ be continuously differentiable and let $\{\boldsymbol{x}^k\}_k$ be generated by the gradient method for solving

$$\min_{\boldsymbol{x}} \ f(\boldsymbol{x}) \quad \text{s.t.} \quad \boldsymbol{x} \in \mathbb{R}^n$$

with one of the following step size strategies:

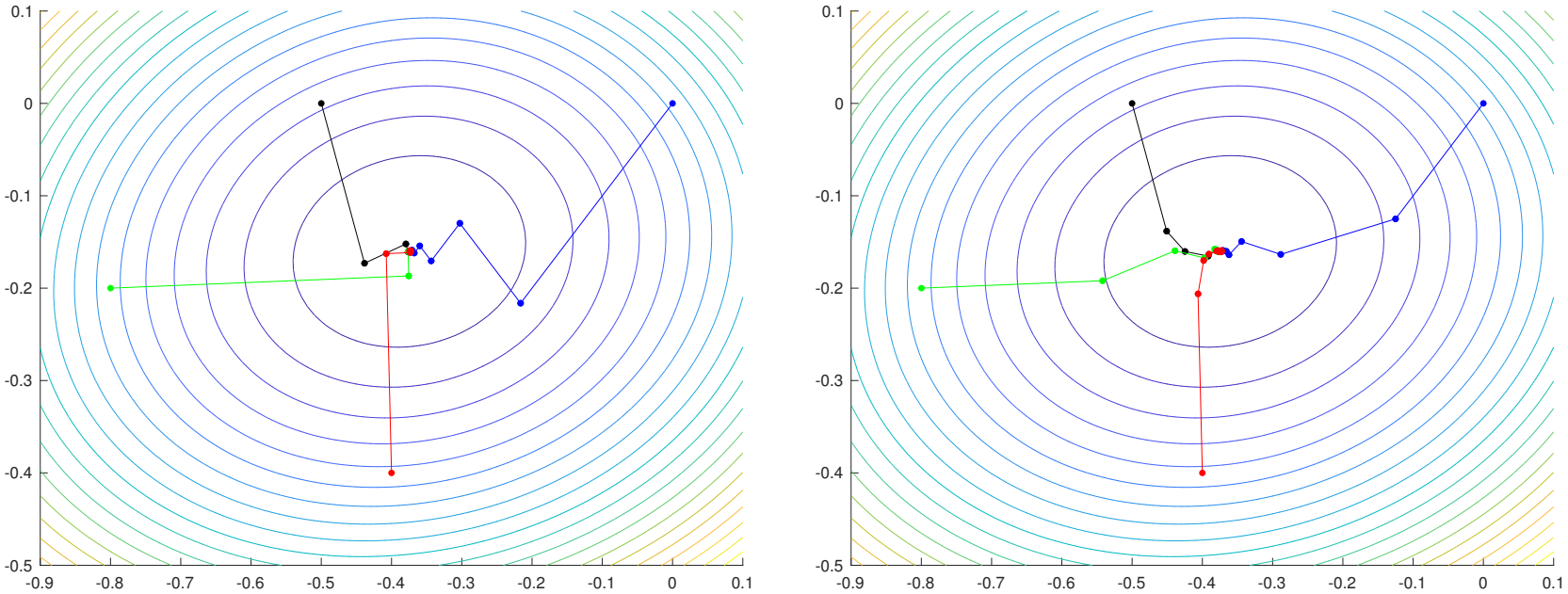
- Exact line search,
- Armijo line search (backtracking) with $\sigma, \gamma \in (0, 1)$.

Then, $\{f(\boldsymbol{x}^k)\}_k$ is nonincreasing and every accumulation point of $\{\boldsymbol{x}^k\}_k$ is a stationary point of f.

Global Convergence: Illustration

We use the same function as example:

$$f(\boldsymbol{x}) = \exp(x_1 + x_2) + x_1^2 + 3x_2^2 - x_1 x_2$$



- **Left:** exact line search. **Right:** backtracking.

Rates of Convergence

Assumption: Lipschitz Continuity

Definition: Lipschitz Continuity and Smoothness

A function f is Lipschitz smooth over Ω if its gradient ∇f is Lipschitz continuous:

$$\|\nabla f(\boldsymbol{x}) - \nabla f(\boldsymbol{y})\| \leq L\|\boldsymbol{x} - \boldsymbol{y}\|, \quad \forall\, \boldsymbol{x}, \boldsymbol{y} \in \Omega,$$

where $L > 0$ is called Lipschitz smoothness constant.

Remark: it means that gradient does not change too fast.

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Definition: Lipschitz Continuity and Smoothness

A function f is Lipschitz smooth over Ω if its gradient ∇f is Lipschitz continuous:

$$\|\nabla f(\boldsymbol{x}) - \nabla f(\boldsymbol{y})\| \leq L\|\boldsymbol{x} - \boldsymbol{y}\|, \quad \forall\, \boldsymbol{x}, \boldsymbol{y} \in \Omega,$$

where $L > 0$ is called Lipschitz smoothness constant.

Theorem: Lipschitz Continuity via Hessians

Let f be twice continuously differentiable. Then, the following two conditions are equivalent:

- f is Lipschitz smooth with constant L.
- $\|\nabla^2 f(\boldsymbol{x})\| \leq L$ for any $\boldsymbol{x} \in \mathbb{R}^n$.

Examples of Lipschitz Smooth Functions

Examples:

- The linear function $f(\boldsymbol{x}) := \boldsymbol{b}^\top \boldsymbol{x} + c$, $\boldsymbol{b} \in \mathbb{R}^n$, $c \in \mathbb{R}$, is Lipschitz smooth with $L = 0$.

① By definition, $\nabla f(x) = b$.

$$\|\nabla f(x) - \nabla f(y)\| = \|b - b\| = 0 = 0\|x - y\|$$

$L = 0$

② Or by the previous theorem

$$\nabla f(x) = b, \quad \nabla^2 f(x) = 0 \leq 0 \implies L = 0$$

Examples of Lipschitz Smooth Functions

Examples:

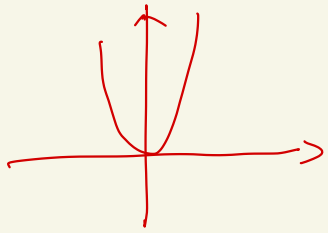
- The linear function $f(\boldsymbol{x}) := \boldsymbol{b}^\top \boldsymbol{x} + c$, $\boldsymbol{b} \in \mathbb{R}^n$, $c \in \mathbb{R}$, is Lipschitz smooth with $L = 0$.
- Consider the quadratic function $f(\boldsymbol{x}) := \frac{1}{2}\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x} + \boldsymbol{b}^\top \boldsymbol{x} + c$:

$$\begin{aligned}\|\nabla f(\boldsymbol{x}) - \nabla f(\boldsymbol{y})\| &= \|(\boldsymbol{A}\boldsymbol{x} + \boldsymbol{b}) - (\boldsymbol{A}\boldsymbol{y} + \boldsymbol{b})\| \\ &= \|\boldsymbol{A}(\boldsymbol{x} - \boldsymbol{y})\| \le \|\boldsymbol{A}\| \cdot \|\boldsymbol{x} - \boldsymbol{y}\|.\end{aligned}$$

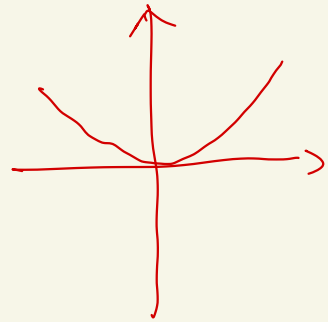
  Hence, f is Lipschitz smooth with $L = \|\boldsymbol{A}\|$. Here, the norm $\|\boldsymbol{A}\|$ denotes the spectral norm of $\boldsymbol{A}$:

$$\|\boldsymbol{A}\| = \sqrt{\lambda_{\max}(\boldsymbol{A}^\top \boldsymbol{A})} = \max_{\|\boldsymbol{d}\|=1} \|\boldsymbol{A}\boldsymbol{d}\|$$

Consider $f(x) = \frac{1}{2} a x^2$



$a = 10$



$a = 1$

$\nabla^2 f(x) = a = 10$

$(L = 10)$

$\nabla^2 f(x) = a = 1$

$(L = 1)$

The smaller L is, the function gradient changes slower, or the function is "smoothier"

Definition: Linear Convergence

We say that $\{\boldsymbol{x}^k\}_k$ converges linearly with rate $\eta \in (0,1)$ to $\boldsymbol{x}^* \in \mathbb{R}^n$ if

$$\|\boldsymbol{x}^{k+1} - \boldsymbol{x}^*\| \leq \eta \cdot \|\boldsymbol{x}^k - \boldsymbol{x}^*\|.$$

$$\Rightarrow \|x^k - x^*\| \leq \eta \|x^{k-1} - x^*\| \leq \eta^2 \|x^{k-2} - x^*\|$$

$$\cdots \leq \eta^k \|x^0 - x^*\|$$

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Remark:

- Sometimes it refers to that the above inequality holds for k large enough: $k \geq \ell$ where $\ell \geq 0$.

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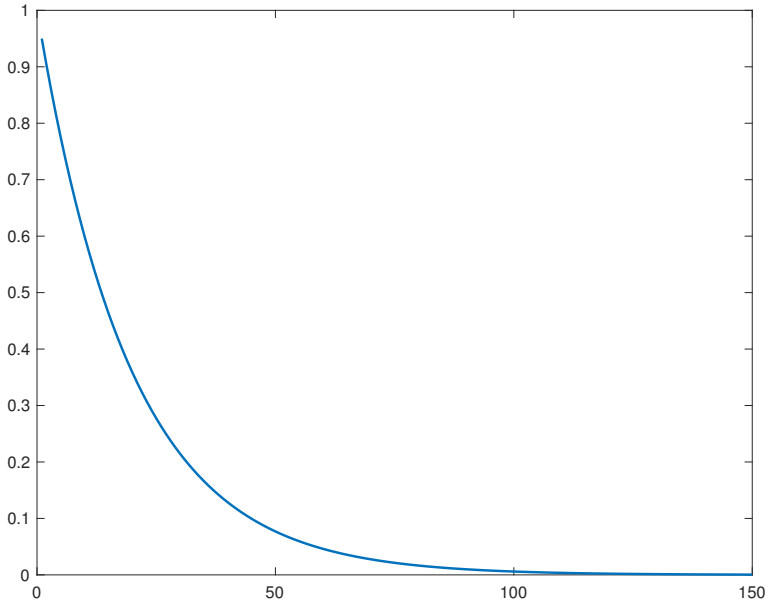
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Example:

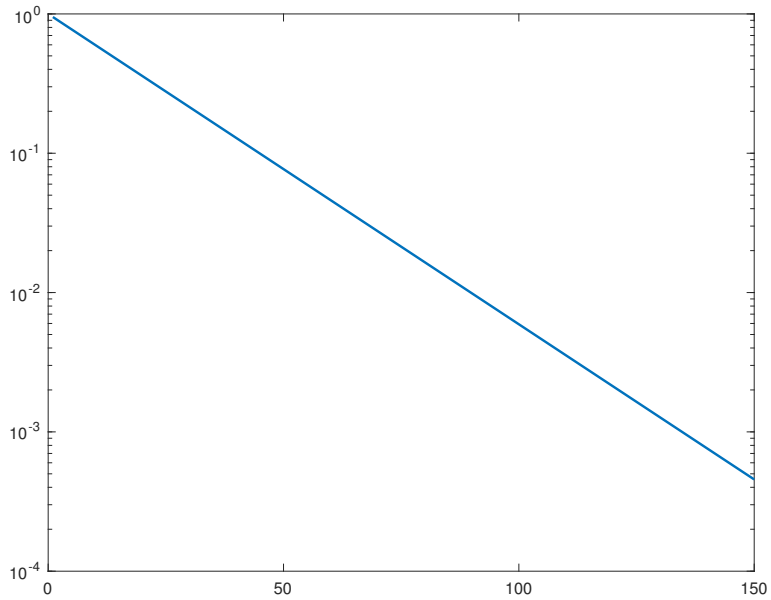
- Let $\rho \in (0,1)$ be given, then the sequence $\{x^k\}_k$ with $x^k := \rho^k$ converges linear to $x^* = 0$ with rate ρ:

$$\frac{|x^{k+1} - x^*|}{|x^k - x^*|} = \frac{\rho^{k+1}}{\rho^k} = \rho, \quad \forall\ k \geq 0.$$

Illustration of Linear Convergence



(a) Plot of $\{0.95^k\}_k$



(b) Logarithmic plot of $\{0.95^k\}_k$

- The plot in (b) shows convergence of the adjusted sequence $\tilde{x}^k = \log_{10}(0.95^k) = \log_{10}(0.95) \cdot k \approx -0.022 \cdot k$.
- The labels of the y-axis are given by $10^{\tilde{x}^k}$.
- In logarithmic plots, linear convergence corresponds to a linear line in k .

Strongly Convex Function

Definition: Strongly Convex Function

A function $f : \Omega \to \mathbb{R}$ is strongly convex with parameter $\mu > 0$ if, for all $\boldsymbol{x}, \boldsymbol{y} \in \Omega$, it satisfies the inequality:

$$f(\boldsymbol{y}) \geq f(\boldsymbol{x}) + \nabla f(\boldsymbol{x})^T(\boldsymbol{y} - \boldsymbol{x}) + \frac{\mu}{2}\|\boldsymbol{y} - \boldsymbol{x}\|^2,$$

where μ is called the strong convexity constant.

$\mu = 0 \longrightarrow$ convex function

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Remark:

- A twice-differentiable function $f : \mathbb{R}^n \to \mathbb{R}$ is strongly convex with parameter $\mu > 0$ if and only if the smallest eigenvalues $\lambda_{\min}(\nabla^2 f(\boldsymbol{x})) \geq \mu$ for any $\boldsymbol{x}$.

Example: $f(x) = \frac{1}{2}x^T A x$ where A is positive definite.

Theorem: Rates for Strongly Convex Problems

Let f be strongly convex with constant μ and Lipschitz smooth with constant L. Let $\{\boldsymbol{x}^k\}_k$ be generated by the gradient descent method and let $\boldsymbol{x}^*$ be the global minimizer of $\min_{\boldsymbol{x}} f(\boldsymbol{x})$. Then with proper constant step size

$$\{\boldsymbol{x}^k\}_k \text{ converges linearly to } \boldsymbol{x}^*.$$

$\Rightarrow \|x^{k+1} - x^*\| \le \rho \|x^k - x^*\|$ for some $\rho \in (0,1)$

Moreover, with proper constant step size, exact line search or Armijo line search, $\{f(\boldsymbol{x}^k)\}_k$ converge linealry to $f(\boldsymbol{x}^*)$ with $\eta := 1 - 2M\mu \in (0,1)$ (see next slide for M):

$$f(\boldsymbol{x}^{k+1}) - f(\boldsymbol{x}^*) \le \eta \cdot [f(\boldsymbol{x}^k) - f(\boldsymbol{x}^*)].$$

Convergence Rate: Remarks

Since $\eta = 1 - 2M\mu$, we want to find $\bar{\alpha} \in (0, \frac{2}{L})$ to maximize $\bar{\alpha}(1 - \frac{L\bar{\alpha}}{2}) \to \bar{\alpha} = \frac{1}{L}$.

Remarks:

- The constant M depends on the chosen line search procedure:

$$M = \begin{cases} \bar{\alpha}(1 - \frac{L\bar{\alpha}}{2}) & \text{constant step size: } \bar{\alpha} \in (0, \frac{2}{L}), \\ \frac{1}{2L} & \text{exact line search,} \\ \gamma \min\{1, \frac{2\sigma(1-\gamma)}{L}\} & \text{Armijo line search.} \end{cases}$$

- The optimal rate is achieved for the constant step size $\bar{\alpha} = \frac{1}{L}$ or exact line search. In this case, we obtain:

A fact: $L \geq \mu$.

$$\eta = 1 - \frac{\mu}{L} = 1 - \frac{1}{\kappa} \quad \text{where} \quad \boxed{\kappa = \frac{L}{\mu}.}$$

called condition number.

- The rate $\bar{\eta}$ for $\{\|\boldsymbol{x}^k - \boldsymbol{x}^*\|\}_k$ typically has the form $\bar{\eta} = \frac{\kappa - 1}{\kappa + 1}$.

Remark: condition number $\kappa \geq 1$. The larger it is, the slower convergence is.

Today's lecture

- Armijo Line Search
- Global Convergence
- Convergence Rate

Next lecture

- Newton's Method

Questions?